Recombination without Exhaustion

Purity of Pure and fundamental propositions

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Can purity itself be pure if there is a fundamental proposition? Under Bacon's full Quantified Logical Necessity, Lettie's theorem answers Goodman's question negatively. His proof uses two instances of zeroary Exhaustion. Shortly after Lettie found his result, a strengthening was found that drops Exhaustion. This note proves it.

The strengthening says that Logical Purity, Application and Purity of Pure at type $t \to t$, taken as ordinary premises over H + PE, already rule out every universal proposition. Since unary Recombination makes a fundamental proposition universal, Purity of Pure is refuted by Logical Purity, Application, unary Recombination and a fundamental proposition. This answers the Recombination-only reading of Goodman's question. No Exhaustion, Persistence or Purity of Fun is used.

1 From fundamentality to a universal proposition

Propositions have type t, $u := t \to t$ is the type of unary propositional operators, and $c := u \to t$. Write $\pi := \mathrm{Pure}_u$, a constant of type c. The instance of Purity of Pure at issue is

$$\mathrm{PP}_u := \mathrm{Pure}_c(\pi).$$

Throughout, p, q, x, y, z, r, b range over propositions and S, X, Y over operators of type u. We use the definitions

$$\top := \forall p(p \to p), \qquad \Box A := (A = \top), \qquad \Diamond A := {\sim}\Box{\sim}A.$$

Following Lettie, put

$$R(x,y) := \forall Y\big(\pi(Y) \to (Y(y) \to \Diamond Y(x))\big), \tag{1}$$

$$\mathrm{Uni}(x) := \forall y\, R(x,y). \tag{2}$$

The relation $R(x,y)$ says that x can have every pure property that y has, where a property is an operator of type u. We call x *universal* if this holds for every proposition y.

Unary Recombination says

$$\mathrm{Rec}_1: \quad \forall X \forall r\big((\mathrm{Pure}_u(X) \,\&\, \mathrm{Fun}_t(r)) \to (\Box X(r) \to \forall z\, X(z))\big).$$

Logical Purity says that closed terms containing no constants are pure. Application at (σ, τ) says that a pure operator applied to a pure argument has a pure value:

$$\mathrm{App}_{\sigma,\tau}: \quad \forall X^{\sigma\to\tau} \forall y^{\sigma} \big((\mathrm{Pure}(X) \,\&\, \mathrm{Pure}(y)) \to \mathrm{Pure}(X(y))\big).$$

Proposition (Lettie: fundamental propositions are universal)**.** *Put* $\mathsf{neg} := \lambda Y \lambda x.{\sim}Y(x)$*. Given* $\mathrm{Pure}(\mathsf{neg})$*,* $\mathrm{App}_{u,u}$ *and* Rec_1*, every fundamental proposition is universal. So if a fundamental proposition exists, a universal one exists.*

Proof. Let r be fundamental. Fix a proposition y and a pure operator Y such that $Y(y)$. The operator $X := \mathsf{neg}\, Y$ is pure, by $\mathrm{Pure}(\mathsf{neg})$, $\mathrm{Pure}_u(Y)$ and $\mathrm{App}_{u,u}$. By β-conversion, ${\sim}Y(r) \to X(r)$ is a theorem, so Necessitation and K give $\Box{\sim}Y(r) \to \Box X(r)$.

If $\Box{\sim}Y(r)$, Recombination at X gives $\forall z\, X(z)$, hence $X(y)$, that is, ${\sim}Y(y)$, contradicting $Y(y)$. Hence $\Diamond Y(r)$. Since y and Y were arbitrary, $\mathrm{Uni}(r)$. The second claim follows by existential instantiation on the fundamental proposition and existential generalization. ■

It therefore suffices to show that PP_u, together with suitable instances of Logical Purity and Application, rules out every universal proposition. The proof of Theorem 1 below contains no Fun.

2 The base logic and the idea of the proof

We work over H + PE: classical higher-order logic H, with Leibniz identity and $\beta\eta$-conversion, closed under modus ponens, Generalization, existential Instantiation, and Propositional Equivalence at type t. The last rule takes a *theorem* $A \leftrightarrow B$ to $A = B$. Modalized Functionality is not assumed. We write C for Bacon's Classicism, which extends H + PE.

The base logic proves the laws

$$\mathrm{K}: \quad \Box(A \to B) \to (\Box A \to \Box B),$$
$$\mathrm{T}: \quad \Box A \to A, \qquad\qquad 4: \quad \Box A \to \Box\Box A,$$

and $\Box(A \leftrightarrow B) \to A = B$. It is closed under Necessitation; hence, by K, a theorem $A \to B$ yields the theorem $\Box A \to \Box B$. We call this the K-rule. These rules apply to base theorems, including theorems that mention the constants Pure_σ. They do *not* authorize necessitating consequences of the purity premises: every premise is ordinary, and "under premises Π, A" means that $\bigwedge \Pi \to A$ is a theorem of H + PE. We mark the points where premises enter.

Where Lettie uses Exhaustion. Lettie calls z *firm* when $F(z) := \Box\big(\sim R(z,\sim z)\,\&\,\sim R(\sim z,z)\big)$. He sets $Q(z) := F(z) \to z$ and introduces two tests, $P(z) := \sim z\,\&\,\sim F(z)$ and $P'(z) := z\,\&\,\sim F(z)$. The heart of his proof is that $\Box\pi(P)$ and $\Box\pi(P')$ make every value of Q firm. The purity premises supply $\pi(P)$ and $\pi(P')$, but only as consequences of premises, so they cannot be necessitated. Zeroary Exhaustion, $\mathrm{Pure}_t(a) \to (a \to \Box a)$, supplies the boxes.

The plan. We relativize Lettie's construction to a *guard* proposition b: in R, the possibility $\Diamond Y(x)$ becomes $\Diamond(b\,\&\,Y(x))$, and firmness becomes $\Box(b \to \cdots)$. The relativized argument needs, in place of $\Box\pi(P)\,\&\,\Box\pi(P')$, only

$$\Box\big(b \to \kappa(b)\big),$$

where $\kappa(b)$ says that the two relativized tests are pure. This can be a theorem of the base logic even though $\kappa(b)$ itself is only a consequence of premises. We therefore look for a proposition a such that

(3) $\quad a$ is pure and true, under the premises, and $\vdash \Box(a \to \kappa(a))$.

Section 3 constructs, for any closed operator Θ, a proposition a such that $a \to \Theta(a)$ is a base theorem. Section 6 shows that the premises make a pure and true whenever Θ is pure and preserves pure truths. Taking $\Theta(b) := b\,\&\,\kappa(b)$ gives (3). Lettie's diagonal then runs relative to a: if r were universal, $e := Q_a(r)$ would satisfy both a separation condition and the relation it forbids.

3 A closure lemma for pure truths

The construction in this section is independent of Recombination and of Lettie's construction. Fix a closed term Θ of type u; it may contain constants. Let $m := \lambda S.\forall p(S(p) \to p)$, so that $m(S)$ says that everything S holds of is true. For X of type u, let $\mathrm{Cl}(X)$ be the conjunction of the following five conditions:

(Cl1) $\quad X(\top)$;

(Cl2) $\quad \forall p\forall q\big((X(p)\,\&\,X(q)) \to X(p\,\&\,q)\big)$;

(Cl3) $\quad \forall p\big(X(p) \to X(\Theta(p))\big)$;

(Cl4) $\quad \forall S\big(\pi(S) \to X(\pi(S))\big)$;

(Cl5) $\quad \forall S\big((\pi(S)\,\&\,\forall p(S(p) \to X(p))) \to X(m(S))\big)$.

Put

$$W := \lambda q.\forall X(\mathrm{Cl}(X) \to X(q)), \qquad a := m(W) \,\&\, \pi(W).$$

Thus W holds of the propositions that belong to every operator satisfying (Cl1)–(Cl5), and a says that W holds only of truths and is pure. With $\Theta := \Phi$, a will be the guard of (3). Two facts about W are used throughout. By *elimination*, $W(q) \to (\mathrm{Cl}(X) \to X(q))$. By *introduction*, from $P \to (\mathrm{Cl}(X) \to X(q))$, with X not free in P, we may infer $P \to W(q)$.

The terms Cl and W mention π and Θ. Abstracting both gives closed terms that contain no constants:

$$\widehat{\mathrm{Cl}} := \lambda c' \lambda T \lambda X.\,((\mathrm{Cl1}) \,\&\, \cdots \,\&\, (\mathrm{Cl5})), \qquad \widehat{W} := \lambda c' \lambda T \lambda q.\forall X(\widehat{\mathrm{Cl}}\, c'\, T\, X \to X(q)),$$

where c' and T stand for π and Θ in the conditions. By β-conversion, $\mathrm{Cl}(X) = \widehat{\mathrm{Cl}}\,\pi\,\Theta\,X$ and $W = \widehat{W}\,\pi\,\Theta$. Likewise m and the conjunction operator $\mathsf{and} := \lambda p \lambda q.(p \,\&\, q)$ contain no constants.

Lemma 1 (W satisfies the closure conditions)**.** $\vdash \mathrm{Cl}(W)$.

Proof. Each condition is proved for W by introduction, from the same condition for an arbitrary X such that $\mathrm{Cl}(X)$. For (Cl1), $\mathrm{Cl}(X)$ gives $X(\top)$, so $W(\top)$. For (Cl2), suppose $W(p) \,\&\, W(q)$ and $\mathrm{Cl}(X)$; elimination gives $X(p)$ and $X(q)$, and (Cl2) for X gives $X(p \,\&\, q)$; now generalize p and q. For (Cl3), elimination turns $W(p)$ into $X(p)$, and (Cl3) for X gives $X(\Theta(p))$. For (Cl4), (Cl4) for X turns $\pi(S)$ into $X(\pi(S))$. For (Cl5), suppose $\pi(S) \,\&\, \forall p(S(p) \to W(p))$ and $\mathrm{Cl}(X)$. For each p, $S(p) \to W(p)$ and elimination give $S(p) \to X(p)$. Generalizing, $\forall p(S(p) \to X(p))$, and (Cl5) for X gives $X(m(S))$. ∎

Lemma 2 ($a \to \Theta(a)$ is a base theorem)**.** $\vdash a \to \Theta(a)$*, and hence* $\vdash \Box(a \to \Theta(a))$.

Proof. Instantiate the conditions of Lemma 1 at W itself. Condition (Cl4) at $S := W$ gives $\pi(W) \to W(\pi(W))$. Condition (Cl5) at $S := W$, with the theorem $\forall p(W(p) \to W(p))$, gives $\pi(W) \to W(m(W))$. Condition (Cl2) gives $(W(m(W)) \,\&\, W(\pi(W))) \to W(a)$, and (Cl3) gives $W(a) \to W(\Theta(a))$. Finally, $m(W) \leftrightarrow \forall p(W(p) \to p)$ by β-conversion, so $m(W) \to (W(\Theta(a)) \to \Theta(a))$.

Now assume a, that is, $m(W)$ and $\pi(W)$. The first three facts give $W(a)$, the fourth gives $W(\Theta(a))$, and the last gives $\Theta(a)$. No premise was used, so $a \to \Theta(a)$ is a theorem of H + PE, and Necessitation applies. ∎

The key condition is (Cl4). At $S = W$ it places the purity claim $\pi(W)$ inside W, and this is what lets $a \to \Theta(a)$ hold in the base logic. Condition (Cl1) is not needed here.

4 The relativized Lettie construction

For a proposition b, define

$$R_b(x,y) := \forall Y\big(\pi(Y) \to (Y(y) \to \Diamond(b \,\&\, Y(x)))\big), \tag{4}$$

$$C_b(z) := {\sim}R_b(z,{\sim}z) \,\&\, {\sim}R_b({\sim}z,z), \tag{5}$$

$$F_b(z) := \Box\big(b \to C_b(z)\big), \tag{6}$$

$$Q_b(z) := F_b(z) \to z, \tag{7}$$

$$P_b(z) := {\sim}z \,\&\, {\sim}F_b(z), \qquad P'_b(z) := z \,\&\, {\sim}F_b(z), \tag{8}$$

$$\kappa(b) := \pi(P_b) \,\&\, \pi(P'_b), \qquad \Phi(b) := b \,\&\, \kappa(b). \tag{9}$$

The relation $R_b(x,y)$ says that x can have, compatibly with b, every pure property that y has. Call z *b-firm* when $F_b(z)$: necessarily, if b, then neither z nor ${\sim}z$ can have every pure property of the other compatibly with b. Up to Boolean identities, the case $b = \top$ is Lettie's construction.

Each of these expressions is built from logical vocabulary, π and b. Abstracting π and b gives closed terms that contain no constants:

$$\begin{aligned}
\hat{R} &:= \lambda c'\lambda b\lambda x\lambda y.\, \forall Y\big(c'(Y) \to (Y(y) \to \Diamond(b \,\&\, Y(x)))\big),\\
\hat{F} &:= \lambda c'\lambda b\lambda z.\, \Box\big(b \to ({\sim}\hat{R}c'bz(\sim z) \,\&\, {\sim}\hat{R}c'b(\sim z)z)\big),\\
\hat{Q} &:= \lambda c'\lambda b\lambda z.(\hat{F}c'bz \to z),\\
\hat{P} &:= \lambda c'\lambda b\lambda z.({\sim}z \,\&\, {\sim}\hat{F}c'bz), \qquad \hat{P}' := \lambda c'\lambda b\lambda z.(z \,\&\, {\sim}\hat{F}c'bz),\\
\hat{\Phi} &:= \lambda c'\lambda b.\big(b \,\&\, (c'(\hat{P}c'b) \,\&\, c'(\hat{P}'c'b))\big).
\end{aligned}$$

By β-conversion, $R_b(x,y) = \hat{R}\pi bxy$, $F_b(z) = \hat{F}\pi bz$, $Q_b = \hat{Q}\pi b$, $P_b = \hat{P}\pi b$, $P'_b = \hat{P}'\pi b$ and $\Phi = \hat{\Phi}\pi$. For R_b we use elimination, its instance at a given Y, and introduction: from $P \to (\pi(Y) \to (Y(y) \to \Diamond(b \,\&\, Y(x))))$, with Y not free in P, infer $P \to R_b(x,y)$.

All results in this section are theorems of H + PE, for arbitrary propositions b, x and z. They mention π and b but make no assumption about what is pure or true.

Lemma 3 (Firmness persists)**.** $F_b(x) \to \Box F_b(x)$.

Proof. By definition $F_b(x)$ is $\Box B$ with $B := b \to C_b(x)$. Axiom 4 gives $\Box B \to \Box\Box B$, and the K-rule applied to $\Box B \to F_b(x)$ gives $\Box\Box B \to \Box F_b(x)$. ∎

Lemma 4 (Q_b fixes firm propositions)**.** $F_b(x) \to Q_b(x) = x$.

Proof. The formula $F_b(x) \to (Q_b(x) \leftrightarrow x)$ is a tautology, given (7). The K-rule gives $\Box F_b(x) \to \Box(Q_b(x) \leftrightarrow x)$. Assume $F_b(x)$. Lemma 3 supplies $\Box F_b(x)$, so $\Box(Q_b(x) \leftrightarrow x)$, and the law $\Box(p \leftrightarrow q) \to p = q$ gives $Q_b(x) = x$. ∎

Lemma 5 (Symmetry)**.** *$F_b(\sim x) \leftrightarrow F_b(x)$, and $P'_b(z) = P_b(\sim z)$.*

Proof. Propositional Equivalence gives $\sim\sim x = x$. By congruence, $R_b(\sim x, \sim\sim x) \leftrightarrow R_b(\sim x, x)$ and $R_b(\sim\sim x, \sim x) \leftrightarrow R_b(x, \sim x)$. Hence $C_b(\sim x) \leftrightarrow C_b(x)$, with the two conjuncts exchanged, and Propositional Equivalence gives $C_b(\sim x) = C_b(x)$. Congruence in $\Box(b \to \cdot)$ gives the first claim.

For the second, $P'_b(z) \leftrightarrow (z \,\&\, \sim F_b(z))$ and $P_b(\sim z) \leftrightarrow (\sim\sim z \,\&\, \sim F_b(\sim z))$. By the first claim the right-hand sides are equivalent, and Propositional Equivalence gives the identity. ∎

The choice $Q_b(x) = F_b(x) \to x$ has a useful consequence: a false value of Q_b is b-firm.

Lemma 6. *$\sim Q_b(x) \to F_b(Q_b(x))$.*

Proof. Assume $\sim Q_b(x)$, that is, $\sim(F_b(x) \to x)$. Then $F_b(x)$, and Lemma 4 gives $Q_b(x) = x$. Substituting $Q_b(x)$ for x in $F_b(x)$ by Leibniz's law yields $F_b(Q_b(x))$. ∎

Lemma 7 (The first test necessarily fails on Q_b's values)**.** *$\Box\sim P_b(Q_b(x))$.*

Proof. Put $q := Q_b(x)$. Then $P_b(q) \leftrightarrow (\sim q \,\&\, \sim F_b(q))$, and Lemma 6 gives $\sim q \to F_b(q)$. Thus $\sim P_b(q)$ is a base theorem, and Necessitation gives the result. ∎

The next lemma is the relativized form of Lettie's key step. The guard appears in it unboxed.

Lemma 8 (The guarded step)**.** *$(b \,\&\, \kappa(b)) \to C_b(Q_b(x))$.*

Proof. Put $q := Q_b(x)$. Assume b, $\pi(P_b)$ and $\pi(P'_b)$, and suppose, for a contradiction, that $C_b(q)$ is false. Thus

$$R_b(q, \sim q) \vee R_b(\sim q, q).$$

The proposition q is not b-firm: $F_b(q)$ would give $b \to C_b(q)$ by T, and so $C_b(q)$. Since $\sim F_b(q)$, the contrapositive of Lemma 6 gives q. Consequently

$$P'_b(q), \qquad P_b(\sim q).$$

The first follows from $q \,\&\, \sim F_b(q)$. For the second, $P_b(\sim q) \leftrightarrow (\sim\sim q \,\&\, \sim F_b(\sim q))$; here q gives $\sim\sim q$, and $F_b(\sim q) \leftrightarrow F_b(q)$ by Lemma 5.

If $R_b(q, \sim q)$, instantiate its quantifier at the pure operator P_b. Since $P_b(\sim q)$, it gives $\Diamond(b \,\&\, P_b(q))$. If instead $R_b(\sim q, q)$, instantiate at the pure operator P'_b. Since $P'_b(q)$, it gives $\Diamond(b \,\&$

$P'_b(\sim q)$). But Lemma 5 at $\sim q$ gives $P'_b(\sim q) = P_b(\sim\sim q)$, and $\sim\sim q = q$ gives $P_b(\sim\sim q) \leftrightarrow P_b(q)$. Hence $P'_b(\sim q) \leftrightarrow P_b(q)$; Propositional Equivalence makes this an identity, and congruence gives $\Diamond(b \,\&\, P'_b(\sim q)) \leftrightarrow \Diamond(b \,\&\, P_b(q))$. Again $\Diamond(b \,\&\, P_b(q))$.

In both cases this contradicts $\Box\sim(b \,\&\, P_b(q))$, which the K-rule obtains from Lemma 7 and the tautology $\sim P_b(q) \to \sim(b \,\&\, P_b(q))$. The remaining reasoning is propositional, so the implication is a theorem of H + PE. ∎

Lemma 9 (Necessary purity under the guard suffices)**.** $\Box\big(b \to \kappa(b)\big) \to F_b(Q_b(x))$.

Proof. Lemma 8 gives the theorem $(b \to \kappa(b)) \to (b \to C_b(Q_b(x)))$. The K-rule gives $\Box(b \to \kappa(b)) \to \Box(b \to C_b(Q_b(x)))$, and the consequent is $F_b(Q_b(x))$. ∎

Lettie's form of this lemma needs $\Box\pi(P)\&\Box\pi(P')$. With the guard, $\Box(b \to \kappa(b))$ suffices, and that is something a base theorem can provide.

5 The twenty purity premises

We now state the finite assumptions. Besides the terms above, the final argument uses two more closed terms that contain no constants:

$$Z := \lambda a \lambda Y \lambda x.(a \,\&\, Y(x)), \qquad \mathsf{comp} := \lambda Y \lambda S \lambda x.\, Y(S(x)).$$

By β-conversion, $Z\,a\,Y\,x \leftrightarrow (a \,\&\, Y(x))$ and $\mathsf{comp}\,Y\,S\,x \leftrightarrow Y(S(x))$. We use ten instances of Logical Purity,

$$\begin{aligned} &\mathrm{L1} : \mathrm{Pure}_t(\top), \quad \mathrm{L2} : \mathrm{Pure}(\mathsf{and}), \quad \mathrm{L3} : \mathrm{Pure}(m), \quad \mathrm{L4} : \mathrm{Pure}(\widehat{W}), \quad \mathrm{L5} : \mathrm{Pure}(\hat{\Phi}), \\ &\mathrm{L6} : \mathrm{Pure}(\hat{P}), \quad \mathrm{L7} : \mathrm{Pure}(\hat{P}'), \quad \mathrm{L8} : \mathrm{Pure}(\hat{Q}), \quad \mathrm{L9} : \mathrm{Pure}(Z), \quad \mathrm{L10} : \mathrm{Pure}(\mathsf{comp}), \end{aligned}$$

and ten instances of Application,

$$\begin{aligned} &\mathrm{A1} : \mathrm{App}_{t,t\to t}, \quad \mathrm{A2} : \mathrm{App}_{t,t}, \quad \mathrm{A3} : \mathrm{App}_{u,t}, \quad \mathrm{A4} : \mathrm{App}_{c,u\to u}, \quad \mathrm{A5} : \mathrm{App}_{u,u}, \\ &\mathrm{A6} : \mathrm{App}_{c,u}, \quad \mathrm{A7} : \mathrm{App}_{c,t\to u}, \quad \mathrm{A8} : \mathrm{App}_{t,u}, \quad \mathrm{A9} : \mathrm{App}_{t,u\to u}, \quad \mathrm{A10} : \mathrm{App}_{u,u\to u}. \end{aligned}$$

Since $u = t \to t$, A1 and A8 are the same formula; it is listed twice because it is used at two different points. Let LA be the conjunction of these premises. None of them is assumed necessary.

Theorem 1. $\mathrm{H} + \mathrm{PE} \vdash \mathrm{LA} \to \big(\mathrm{PP}_u \to \sim\exists r\,\mathrm{Uni}(r)\big)$. *Hence, over C,* PP_u *with the Logical Purity and Application schemas, all as ordinary premises, refutes the existence of a universal proposition.*

We prove the theorem in two stages: the purity premises produce the guard a of (3), and universality then yields a contradiction relative to a.

6 The purity argument

From now on, the premises are LA and PP_u. Put

$$G := \lambda p.(\mathrm{Pure}_t(p) \,\&\, p), \qquad \mathrm{Pres}(\Theta) := \forall p\big((\mathrm{Pure}_t(p) \,\&\, p) \to \Theta(p)\big).$$

Thus $\mathrm{Pres}(\Theta)$ says that Θ holds of every pure truth. The predicate G contains the constant Pure_t. It is used only to instantiate the quantifier in W; its purity is never asserted.

Lemma 10 (The guard is pure and true)**.** *Under L1–L4, A1–A5 and* PP_u*: if* Θ *is pure and* $\mathrm{Pres}(\Theta)$*, then* a *is pure and true.*

Proof. Assume $\mathrm{Pure}_u(\Theta)$ and $\mathrm{Pres}(\Theta)$. We first show $\mathrm{Cl}(G)$, using $G(p) \leftrightarrow (\mathrm{Pure}_t(p) \,\&\, p)$.

- (Cl1): $\top$ is true, and pure by L1.
- (Cl2): if $G(p)$ and $G(q)$, then $p \,\&\, q$ is true. By L2, A1 and A2, $\mathsf{and}\, p\, q$ is pure, and it is $p \,\&\, q$ by β-conversion.
- (Cl3): if $G(p)$, then $\mathrm{Pres}(\Theta)$ gives $\Theta(p)$, and $\mathrm{Pure}_u(\Theta)$, $\mathrm{Pure}_t(p)$ and A2 make $\Theta(p)$ pure.
- (Cl4): if $\pi(S)$, that is, $\mathrm{Pure}_u(S)$, then PP_u and A3 make $\pi(S)$ pure, and $\pi(S)$ is true.
- (Cl5): if $\pi(S)$ and $\forall p(S(p) \to G(p))$, then, since $G(p) \to p$, $\forall p(S(p) \to p)$, which is $m(S)$. L3, $\mathrm{Pure}_u(S)$ and A3 make $m(S)$ pure.

Elimination for W at $X := G$ now gives $W(p) \to G(p)$, hence $W(p) \to p$; generalizing p, we obtain $m(W)$. Next, PP_u, L4 and A4 make $\widehat{W}\pi$ pure, and $\mathrm{Pure}_u(\Theta)$ with A5 makes $W = \widehat{W}\pi\Theta$ pure, which is $\pi(W)$. So $a = m(W) \,\&\, \pi(W)$ is true.

Finally, L3, $\pi(W)$ and A3 make $m(W)$ pure; PP_u, $\pi(W)$ and A3 make $\pi(W)$ pure; and L2, A1 and A2 make their conjunction a pure. ∎

Lemma 11 (Φ is pure and preserves pure truths)**.** *Under the premises,* $\mathrm{Pure}_u(\Phi)$ *and* $\mathrm{Pres}(\Phi)$*.*

Proof. PP_u, L5 and A6 make $\Phi = \hat{\Phi}\pi$ pure. Now let b be pure and true. PP_u, L6 and A7 make $\hat{P}\pi$ pure, and $\mathrm{Pure}_t(b)$ with A8 makes $P_b = \hat{P}\pi b$ pure. This is $\pi(P_b)$. Likewise PP_u, L7, A7, $\mathrm{Pure}_t(b)$ and A8 give $\pi(P'_b)$. So $\kappa(b)$ holds, and with b, $\Phi(b)$. ∎

Now take $\Theta := \Phi$ in Section 3. This fixes $W = \widehat{W}\pi\Phi$ and $a = m(W)\,\&\,\pi(W)$.

Lemma 12 (Properties of the guard)**.** *(a) Under the premises, a is pure and true.*

(b) $\vdash \Box(a \to \kappa(a))$.

(c) $\vdash F_a(Q_a(x))$ *for every* x.

Proof. Part (a) is Lemma 10 with Lemma 11. For (b), $(a \to \Phi(a)) \to (a \to \kappa(a))$ is a theorem, by (9); the K-rule and Lemma 2 give $\Box(a \to \kappa(a))$. For (c), apply Lemma 9 at $b := a$ and use (b). ∎

The premises enter only in (a). Parts (b) and (c) are theorems of the base logic, so no purity or truth claim has been necessitated. This is (3).

7 The contradiction

Lemma 13 (Universality relative to the guard)**.** *Under the premises, if r is universal, then $R_a(r, y)$ for every y.*

Proof. Let Y be pure with $Y(y)$, and put $Z_Y := Z\,a\,Y$. L9, $\mathrm{Pure}_t(a)$ and A9 make $Z\,a$ pure, and $\mathrm{Pure}_u(Y)$ with A5 makes Z_Y pure. By β-conversion, $Z_Y(y)$ follows from a, which is true, and $Y(y)$. Universality of r gives $R(r, y)$, and its instance at Z_Y gives $\Diamond Z_Y(r)$. Since $\sim(a\,\&\,Y(r)) \to \sim Z_Y(r)$ is a theorem, the K-rule turns $\Diamond Z_Y(r)$ into $\Diamond(a\,\&\,Y(r))$. Introduction for R_a gives $R_a(r, y)$. ∎

Proof of Theorem 1. Assume the premises, and suppose that r is universal. Put $e := Q_a(r)$. We first show $R_a(e, \sim e)$. Let Y be pure with $Y(\sim e)$, and put $Y' := \mathsf{comp}\,Y\,Q_a$.

The operator $Q_a = \hat{Q}\pi a$ is pure: PP_u, L8 and A7 make $\hat{Q}\pi$ pure, and $\mathrm{Pure}_t(a)$ with A8 makes Q_a pure. Hence Y' is pure: L10, $\mathrm{Pure}_u(Y)$ and A10 make $\mathsf{comp}\,Y$ pure, and $\mathrm{Pure}_u(Q_a)$ with A5 makes Y' pure.

Next, $Y'(\sim e)$. Lemma 12(c) at $x := r$ makes e a-firm. By symmetry $\sim e$ is a-firm too, so Lemma 4 gives $Q_a(\sim e) = \sim e$. Leibniz's law turns $Y(\sim e)$ into $Y(Q_a(\sim e))$, which is $Y'(\sim e)$.

Lemma 13 at $y := \sim e$ gives $R_a(r, \sim e)$. Its instance at Y' gives $\Diamond(a\,\&\,Y'(r))$. Since $Y'(r) \leftrightarrow Y(e)$, the K-rule gives $\Diamond(a\,\&\,Y(e))$. Introduction for R_a now yields

$$R_a(e, \sim e).$$

But e is a-firm: $\Box(a \to C_a(e))$. By T and the truth of a, $C_a(e)$ holds, and its first conjunct is $\sim R_a(e, \sim e)$. This contradiction shows that, under the premises, no r is universal;

existential Instantiation gives $\sim\exists r\,\mathrm{Uni}(r)$. Separating PP_u from the other premises gives the displayed theorem.

For the form over C, every theorem of H + PE is a theorem of C, and each conjunct of LA is an instance of Logical Purity or Application. Modus ponens with the premises gives the conclusion. ■

Where Lettie's proof uses Exhaustion to necessitate the true purity claims $\pi(P)$ and $\pi(P')$, this proof necessitates only the base theorem $a \to \Phi(a)$. The premises make the guard a pure and true. Its purity lets the operators Z_Y and Q_a serve as pure properties, and its truth discharges the guard after T.

Corollary. *Let* $\mathrm{EF} := \exists r\,\mathrm{Fun}_t(r)$*, and let* $\mathrm{H5} := \mathrm{Pure}(\mathsf{neg}) \,\&\, \mathrm{App}_{u,u} \,\&\, \mathrm{Rec}_1$ *be the hypotheses of the Proposition of Section 1. Then*

$$\mathrm{H} + \mathrm{PE} \vdash \mathrm{LA} \to (\mathrm{EF} \to (\mathrm{H5} \to \sim\mathrm{PP}_u)).$$

In particular, Logical Purity, Application, a unique fundamental proposition, no fundamental entities at other types, and zeroary and unary Recombination, as ordinary premises over C, refute PP_u*.*

Proof. Under EF and H5, the Proposition of Section 1 gives a universal proposition, while LA and PP_u rule one out by Theorem 1. A unique fundamental proposition gives EF. For the second claim, the premises used are Logical Purity (L1–L10 and $\mathrm{Pure}(\mathsf{neg})$), Application, a unique fundamental proposition and Rec_1. The absence of fundamental entities at other types and zeroary Recombination are not used. Adding PP_u to these premises therefore derives a contradiction. ■